

# Path Parameters, Static Files, and Debugging

Section Review

# Path Parameters I

- A **path parameter** is a dynamic, named segment of the URL/path.
- Wrap the path parameter in curly braces in the route string (e.g., **/coffees/{coffee\_id}**)
- FastAPI will supply the path value as a parameter to the route handler function. Give the function parameter the same name as the path parameter.

# Path Parameters II

- Declare the type of the function parameter.
- Path parameters arrive as strings but FastAPI can automatically convert to a different data type.
- For example, **coffee\_id: int** will attempt a conversion to an integer type.
- If conversion fails, FastAPI will return an error response.

# The 404 Status Code

- The **404 Not Found** status code indicates the server was unable to find the requested resource.
- A function without an explicit return value will implicitly return **None**, which JSON will convert into **null**.
- The status code will still be 200 OK.

# The 404 Status Code II

- Import the **HTTPException** error class from **fastapi**.
- Raise an instance of the **HTTPException** class and pass in **status\_code** and **detail** parameters.
- For **status\_code**, you can refer to the right constant from the **status** module.
- The **detail** parameter will be the error sent back to the client.

# Route Order

- FastAPI matches routes from top to bottom, in the order they are declared/registered.
- A route with a path parameter *may* match another route. For example, **/coffees/milk** will match **/coffees/{coffee\_id}**.
- Declare the routes with fixed segments before the route with path parameters.



# Static Files

- FastAPI can serve static files (files from a directory).
- FastAPI will create a route for every filename. Nested directories become `/` in the route name.
- Invoke **app.mount**, provide a prefix for the path, and pass an instance of the **StaticFiles** class.
- Static files should only be public files. Do not place any sensitive files in the directory.

# Debugging in VSCode

- Configure a **launch.json** file to launch the FastAPI/Uvicorn server from within VSCode.
- Place a breakpoint by clicking on the gutter to the left of the line number.
- Use the toolbar to jump to the next breakpoint, progress to the next line, step into a function, or step out of a function.